AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2011

General Mathematics Paper II

Time allowed: 2 hours 20 minutes Marks 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are THIRTEEN questions. Answer TEN questions. Choices are specified inside the paper

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use colour pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

(ATTEMPT EITHER Q.1 OR Q.2)

Q.1. (Total 5 Marks)

Simplify $\frac{27xy^3 - 3x^3y}{(3y - x)^2} \div \frac{3xy}{3y - x}$.

(ATTEMPT EITHER Q.1 OR Q.2)

Q.2. (Total 5 Marks)

a. Prove that $(a^2 + ab + b^2)(a^2 - ab + b^2)(a + b)(a - b) = a^6 - b^6$. (4 Marks)

b. Find the value of $1 - a^3$ if $a = -2$. (1 Mark)

(ATTEMPT EITHER Q.3 OR Q.4)

Q.3. (Total 5 Marks)

Factorize the following completely:

i. $(x+1)^2 - 49$ (2 Marks)

ii. $16ax^2 - 16ax + 4a$ (3 Marks)

(ATTEMPT EITHER Q.3 OR Q.4)

Q.4. (Total 5 Marks)

If 2 is a zero of polynomial $P(x) = x^3 - 3x - 2$, then completely factorize $P(x)$.

(ATTEMPT EITHER PART a OR PART b OF Q.5.)

Q.5. (Total 4 Marks)

a. Calculate the square root of the algebraic expression $1-6x+11x^2-6x^3+x^4$.

b. Find the highest common factor (H.C.F) of $(x^2-4),(x^3+8)$ and $(x+2)$ by factorization method.

Q.6. (Total 5 Marks)

a. Find the value of x for $\frac{3-x}{2}+\frac{6-x}{3}=0$. (3 Marks)

b. Find the solution set of $|x-1|=1$. (2 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.7.)

Q.7. (Total 5 Marks)

a. Find the solution set of $n^2 - n + \frac{2}{9} = 0$ by the factorization method.

b. Ms. Rana bought *x* number of file covers for an amount of rupees $x^2 - 39$. The cost of each file cover is Rs. 10. Find the number of file covers she bought.

(ATTEMPT EITHER Q.8 OR Q.9)

Q.8. (Total 5 Marks)

Find multiplicative inverse $\left(A^{-1}\right)$ of matrix $A = \begin{bmatrix} 3 & -5 \\ 1 & -2 \end{bmatrix}$ and prove that $AA^{-1} = I$.

(ATTEMPT EITHER Q.8 OR Q.9)

Q.9. (Total 5 Marks)

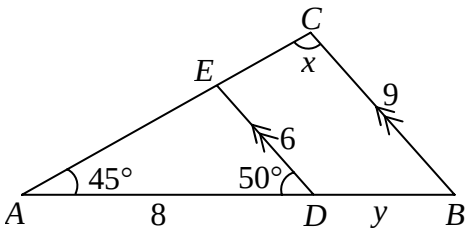
With the help of matrices $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$,

i. demonstrate that the commutative law holds true for addition.

ii. demonstrate that the commutative law does **NOT** hold true for multiplication.

Q.10. (Total 5 Marks)

In the following diagram *BC* is parallel to *DE*. Find the values of *x* and *y*.

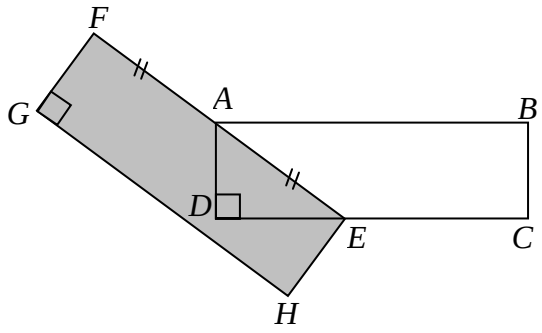


Q.11. (Total 4 Marks)

Construct a triangle *XYZ* with $m\overline{XY} = 6\text{cm}$, $m\angle X = 60°$ and $m\angle Y = 50°$. Also draw two medians of the triangle *XYZ*.

Q.12. (Total 4 Marks)

If two congruent rectangles *ABCD* and *EFGH* are partially overlapped as shown in the figure, find the length of *DE*, when $mAB = 20\,\text{cm}, GF = 9\,\text{cm}$ and $FA = AE$.



Q.13. (Total 3 Marks)

Find the shortest distance between points $A(2,-2)$ and $B(-2,2)$.

END OF PAPER

Please use this page for rough work

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